Arthur Wang

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WORK EXPERIENCE

GIGABYTE Technology
Jun. 2024 – Present

AI Algorithm Deputy Manager / Technical Lead
New Taipei, Taiwan

- Led development of **6+ on-device and enterprise AI applications** spanning local agents, hardware Q&A, translation, and meeting intelligence.
- Architected offline LLM systems with **llama.cpp, vLLM, LangChain, and LlamaIndex**, delivering production AI features under **4GB VRAM**.
- Built RAG and knowledge graph pipelines over **1,000+ product documents**, cutting FAE lookup time from **5–10 minutes to under 30 seconds**.
- Developed reliable Agent workflows with planning, memory, tool invocation, scheduling, permission control, durable execution, and recovery.
- Collaborated with product, engineering, FAE, customer service, and management teams to ship AI capabilities into laptop-side applications, internal support systems, and hardware product workflows.

Tarobo Investment
Jun. 2023 – Oct. 2023

Quantitative Trading Engineer
Taipei, Taiwan

- Developed deep learning models for stock price direction prediction, improving directional prediction accuracy by **15%**, reducing strategy evaluation time from **6 hours to 15 minutes**.
- Analyzed market signals, financial indicators, and model outputs to evaluate trading opportunities and improve strategy iteration.

KEY PROJECTS

GiMATE Agent: On-Device Agentic AI Assistant for Windows
Feb. 2026 – Present

- **Led end-to-end development of a Windows AI desktop assistant**, building the **Electron frontend and FastAPI backend**, integrating local LLM inference, and packaging the application into a standalone **exe** powered by **Qwen3.5-4B**.
- **Built an AI Agent ecosystem with 22 skills and 76 tools**, enabling autonomous workflows across **PowerPoint generation, data analysis, local-file RAG, hardware control, Web Search**, and third-party integrations including **Gmail, Notion, and Outlook**.
- **Designed a small-model Agent architecture** combining **Plan-and-Execute, progressive skill discovery, and deterministic tool execution**, reducing unnecessary LLM calls while enabling reliable multi-step automation on resource-constrained devices.
- **Developed and evaluated a self-correcting execution engine** with dynamic replanning, SQLite task persistence, cross-session memory, and human approval, achieving **97% routing accuracy across 385 test cases**.

GRACE-Mem: Long-Term Memory for Conversational Agents

Oct. 2025 – Jul. 2026

- Architected a **source-evidence-centric long-term memory system** that decouples graph-based candidate discovery, compressed-record reranking, and provenance-linked raw-dialogue reconstruction, preventing lossy summaries from becoming final answering evidence.
- Designed a hybrid retrieval stack combining **BM25 and dense entity/relation search, two-hop query-conditioned spreading activation, and hierarchical reranking**. Instruction-conditioned reranking increased gold-evidence recall from **34.5% to 63.4% (+28.9 points)** and answer accuracy by **3.0 points**.
- Built a production-safe **Evidence Curation Agent** with a bounded 10-call GREP / READ / VEC loop, trajectory-level provenance validation, additive coverage recovery, and identity fallback to recover missing evidence without introducing unsupported source IDs.
- Achieved up to **86.9% on LoCoMo** and **80.2% on LongMemEval**; the GPT-4o-mini configuration surpassed the previously reported state of the art, AMA, by **7.9 / 3.4 points**.
- Ran component-level ablations showing graph retrieval contributed **17.6 points** on LoCoMo, while reconstructing provenance-linked raw dialogue instead of answering from compressed records added **7.8 / 6.2 points**.

FAE Support System: Hybrid RAG for Hardware Q&A

Jan. 2026 – Present

- Built a Hybrid RAG hardware Q&A system covering GIGABYTE product lines, supporting specification lookup, compatibility checks, and troubleshooting for motherboards, GPUs, and laptops.
- Established a spec-to-structured-data pipeline parsing and standardizing **1,000+ product documents** into a Neo4j hardware knowledge graph.
- Deployed BERT-based query classification and entity linking to identify product models, categories, and ambiguous hardware references from user questions.
- Combined graph-based retrieval, BM25, embeddings, and fuzzy matching to support exact model queries, fuzzy product descriptions, and natural-language troubleshooting.
- Achieved approximately **95% model identification accuracy**, **97% Top-3 entity linking hit rate**, and **92% hardware Q&A accuracy** on internal test sets.
- Reduced FAE specification and compatibility lookup time from **5–10 minutes to under 30 seconds**, cutting manual search effort by approximately **70%**.

SIDE PROJECTS

StockRadar — Autonomous Equity Research & Trading Agent

Feb. 2025 – Present

- **Built a 24/7 autonomous AI equity analyst** that continuously identifies investment opportunities and generates actionable buy, add, hold, or short recommendations, complete with entry ranges, price targets, stop-loss levels, and risk/reward ratios.
- **Designed an agentic research workflow** that combines quantitative stock screening with autonomous tool calling. The AI Agent cross-analyzes news, earnings, technical charts, and DCF valuations while incorporating market conditions and portfolio exposure to deliver evidence-backed investment decisions.
- **Developed an automated investment research platform** covering **18 sectors, 60+ financial metrics, and 32 technical indicators**. The system tracks historical research, detects emerging investment opportunities, and delivers real-time alerts via Telegram.

Alpha Note — Autonomous Research & Knowledge Intelligence Agent

Jun. 2026 – Present

- **Built an autonomous AI research analyst** that monitors **90+ technology sources** across AI/ML, software engineering, cybersecurity, arXiv papers, trending GitHub repositories, and Hugging Face models. It selects the **Top 3 must-read stories** and identifies **2–3 cross-source trends**, prioritizing actionable insights over simple summaries.
- **Designed a multi-stage agentic research pipeline** that follows a **Screening** $\rightarrow$ **Deep Research** $\rightarrow$ **Cross-Source Synthesis** workflow. The Agent evaluates relevance, quality, and timeliness, analyzes high-value content, and compares agreements and conflicting perspectives to generate evidence-backed research reports.
- **Built an automated Podcast and YouTube research pipeline** using **Whisper ASR and LLM long-context analysis** to transform hours of content into structured research memos. Extracted key insights, technical concepts, investment implications, and risks are linked to original sources and timestamps, then automatically organized into a searchable **Obsidian knowledge base**.

PUBLICATIONS

GRACE-Mem: Graph Retrieval with Agentic Curation of Evidence for Long-Term Conversational Memory

- **Heng (Arthur) Wang** (first author), X.-N. Chen, P.-Y. Chang, C.-H. Chen, H.-Y. Huang, C. Lin, I-L. Chung. *EMNLP 2026 Workshop on Automated Knowledge Base Construction (AKBC)*, 2026. **Featured Paper (3 selected); Best Paper Award Candidate.**

EDUCATION

National Chiao Tung University Sep. 2017 – Jun. 2021
Bachelor of Science in Computer Science Hsinchu, Taiwan

PROFESSIONAL SKILLS

Agentic AI & Retrieval : Agent Harness Engineering, MCP,Skills, Context Engineering, Long-Term Memory, Agentic RAG, GraphRAG, Hybrid Search, Reranking, Agent Evaluation.
Models & Frameworks : llama.cpp, vLLM, LangChain, LlamaIndex, PyTorch, Hugging Face, Quantization, QLoRA, GRPO, OCR, ASR, VLM
Engineering : Python, React, FastAPI, Electron, REST APIs, SQLite
Languages :Mandarin Chinese, English